

NARMADA SAHOO

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Technical Skills

Languages and Databases: Python, Java, MySQL, MongoDB

AI/ML & Frameworks: PyTorch, Hugging Face Transformers, LangGraph, RAG

Cloud & Engineering: AWS, AWS Bedrock Knowledge Bases, FastAPI, Git, Linux, REST APIs, WebSocket, STT, TTS

Experience

RND OptimizAR

Coimbatore, India

AI Engineer

06/2025-Present

- Built and optimized RAG pipelines using **AWS Bedrock Knowledge Bases**, improving retrieval quality to provide relevant domain and system context to downstream LLM applications.
- Developed and deployed **real-time voice agents** with state management, interruption handling (barge-in), context retention, low-latency responses, and LLM-driven conversational orchestration.
- Designed and implemented **multi-step document processing pipelines using LangGraph** for medical information extraction, reducing processing time by 60% through parallel workflow orchestration and intelligent task routing.
- Served as AI team POC, translating business requirements into scalable AI solutions, selecting appropriate hardware (GPUs), and optimizing computational resource usage.

AI Engineer Intern

01/2025-05/2025

- Developed a medical document classification system by fine-tuning transformer models to enhance accuracy.
- Applied Optical Character Recognition (OCR) and LLM-based question-answering techniques to extract structured and unstructured data from scanned medical documents, improving recognition accuracy through advanced image processing and reducing search time for relevant information by 50%.

Tata Advanced System

Hyderabad, India

Business Development Executive

11/2020-08/2022

- Developed an automated information extraction system using Python and Regex to extract Bill of Materials (BOM) data from PDF documents, significantly reducing manual effort by 70%.
- Served as Aerostructure Assembly Estimator and Configuration Manager for the Business Development Estimation Team, reviewing engineering dossiers to identify and highlight areas of concern needing clarification from customers.

Education

National Institute of Technology, Tiruchirappalli

Tamil Nadu, India

Master of Technology, Data Analytics (CGPA: 9.02/10)

08/2023 - 06/2025

Odisha University of Technology and Research, Bhubaneswar

Odisha, India

Bachelor of Technology, Mechanical Engineering (CGPA: 8.90/10)

08/2016 - 08/2020

Project

Personalized Message Notification Router - Hacker Rank Orchestrate

- Built a personalized notification-routing system using **behavioral-precedent retrieval and TF-IDF similarity** to classify messages into notify, digest, or mute actions.
- Implemented **LLM adjudication, deterministic safety rules, OCR/ASR processing, and caching** for robust and context-aware message routing.

Predicting Emotional Causes Through Question Answering

- Developed a two-stage **BERT-based NLP pipeline** for emotion classification and extraction of emotion-triggering spans using question answering.
- Applied **SHAP and LIME** for model explainability highlighting emotion-triggering words achieving **0.97 F1 for classification and 0.78 F1 for QA** on the available dataset

Achievements

- Ranked **3rd** in M.Tech Data Analytics at NIT Tiruchirappalli, with a CGPA of **9.02/10**
- Secured **347th rank** among **1,978** participants in the Hacker Rank August-2026 Orchestrate challenge.